Special Summer Integrated Learning Programme

Mathematics

Polynomial and Rational Functions

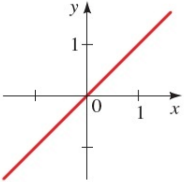
Week 2 Day 1 PM

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

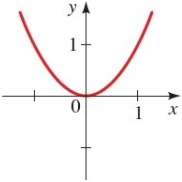
Outline

Graphs of x^n

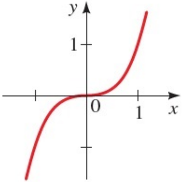
The graph of $P(x) = x^n$ has the same general shape as the graph of $y = x^2$ (resp. $y = x^3$) when n is even (resp. odd). However, as the degree n becomes larger, the graphs become flatter around the origin and steeper elsewhere.



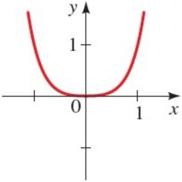
(a) $y = x$



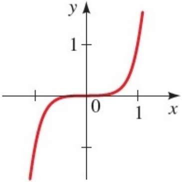
(b) $y = x^2$



(c) $y = x^3$



(d) $y = x^4$



(e) $y = x^5$

End behavior

The greater the degree and the number of terms present in a polynomial, the more complicated its graph can be. However, the graph of a polynomial function has no breaks or holes. Moreover, the graph of a polynomial function is a smooth curve; that is, it has no corners or sharp points.

Since the domain of a polynomial is $\mathbb{R}$, we want to study its ***end behavior***, i.e. what happens to its value when x becomes arbitrarily large in the positive or negative direction. For this, we define the following notations for convenience.

Definition:

- We say that x ***goes to infinity***, denoted as $x \to \infty$, when x increases without bound, i.e. x gets arbitrarily large.
- We say that x ***goes to negative infinity***, denoted as $x \to -\infty$, when x decreases without bound, i.e. x gets arbitrarily negatively large.

End behavior - example

E.g.: The polynomial $y = x^2$ has the following end behavior.

$$y \to \infty \text{ as } x \to -\infty \text{ and } y \to \infty \text{ as } x \to \infty.$$

We also denote the above as

$$\lim_{x \to -\infty} x^2 = \infty \text{ and } \lim_{x \to \infty} x^2 = \infty.$$

E.g.: The polynomial $y = x^3$ has the following end behavior.

$$y \to -\infty \text{ as } x \to -\infty \text{ and } y \to \infty \text{ as } x \to \infty.$$

We also denote the above as

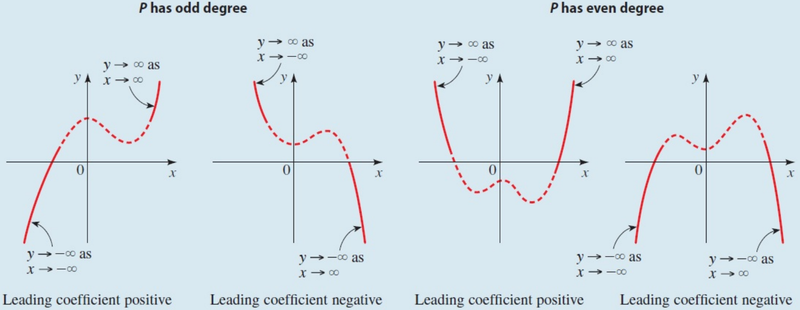
$$\lim_{x \to -\infty} x^3 = -\infty \text{ and } \lim_{x \to \infty} x^3 = \infty.$$

END BEHAVIOR

The end behavior of a polynomial is determined by the term with the highest power of x, because when x is large, the other terms are relatively insignificant in size.

END BEHAVIOR OF POLYNOMIALS

The end behavior of the polynomial $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is determined by the degree n and the sign of the leading coefficient a_n, as indicated in the following graphs.

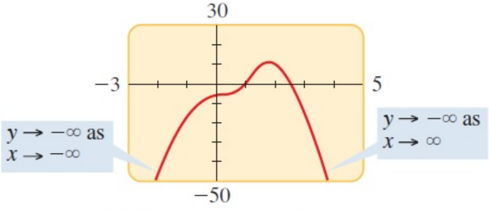


End behavior - example

E.g.: Consider the polynomial $P(x) = -2x^4 + 5x^3 + 4x - 7$. The term that dominates when $x \to \pm\infty$ is $-2x^4$, which has a negative coefficient and an even power. So when $x \to \pm\infty$, $-2x^4 \to -\infty$, and thus

$$P(x) \to -\infty \quad \text{as} \quad x \to -\infty \quad \text{or} \quad x \to \infty.$$

We can use this information to aid us in sketching the end behavior of the graph of $P(x)$.



$P(x) = -2x^4 + 5x^3 + 4x - 7$

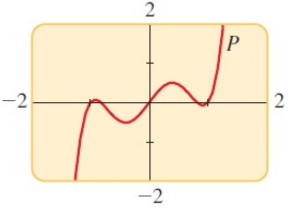
As for the shape of the graph in the middle (i.e. x is not large), we need differential calculus to help us to sketch it.

End behavior - example

E.g.: Consider the polynomial $P(x) = 3x^5 - 5x^3 + 2x$. The term that dominates when $x \to \pm\infty$ is $3x^5$, which has a positive coefficient and an odd power. So when $x \to -\infty$, $3x^5 \to -\infty$ and when $x \to \infty$, $3x^5 \to \infty$. So

$$P(x) \to -\infty \quad \text{as} \quad x \to -\infty \quad \text{and } P(x) \to \infty \quad \text{as} \quad x \to \infty.$$

We can use this information to aid us in sketching the end behavior of the graph of $P(x)$.



$P(x) = 3x^5 - 5x^3 + 2x$

Activity 1

Find out the end behavior of the following polynomials.

(a) $P(x) = x^4 + x^3 - 16x^2 - 4x + 48$

(b) $Q(x) = -x(x-3)(x+2)$

(c) $R(x) = -(x+1)^2(x-1)^3(x-2)$

Activity 1: solution

(a) Since the leading term of $P(x)$ is x^4, which has a positive coefficient and an even power, we conclude that

$$P(x) \to \infty \text{ as } x \to -\infty \text{ and } P(x) \to \infty \text{ as } x \to \infty.$$

(b) Since the leading term of $Q(x)$ is $-x^3$, which has a negative coefficient and an odd power, we conclude that

$$Q(x) \to \infty \text{ as } x \to -\infty \text{ and } Q(x) \to -\infty \text{ as } x \to \infty.$$

(c) Since the leading term of $R(x)$ is $-x^6$, which has a negative coefficient and an even power, we conclude that

$$R(x) \to -\infty \text{ as } x \to -\infty \text{ and } R(x) \to -\infty \text{ as } x \to \infty.$$

Using zeros to graph polynomials

The real roots/zeros of a polynomial are exactly the x-intercepts of its graph. We can use this information to help us in graphing polynomials.

The Fundamental Theorem of Algebra

A degree n polynomial $P(x)$ can be completely factorize into the following form:

$$P(x) = a_n(x - c_1)^{m_1}(x - c_2)^{m_2} \cdots (x - c_k)^{m_k},$$

where $c_1, c_2, \ldots, c_k$ are distinct roots (possibly non-real complex numbers) of $P(x)$, with $m_1 + m_2 + \cdots + m_k = n$. m_i is called the ***multiplicity*** of the root c_i for the polynomial $P(x)$.

The multiplicity of a real root tells us the shape of the graph near that root.

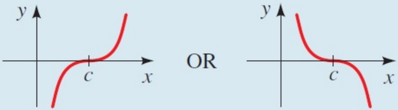
Shape of the graph near a zero

SHAPE OF THE GRAPH NEAR A ZERO OF MULTIPLICITY *m*

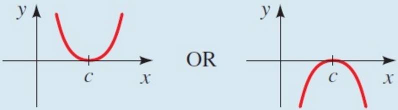
If c is a zero of P of multiplicity m, then the shape of the graph of P near c is as follows.

Multiplicity of *c* | **Shape of the graph of *P* near the *x*-intercept *c***

m odd, $m > 1$



m even, $m > 1$



For $m = 1$, the graph cuts through the x-axis at $x = c$ similar to what a straight line does.

Using zeros to graph polynomials - example

E.g.: Suppose we want to graph

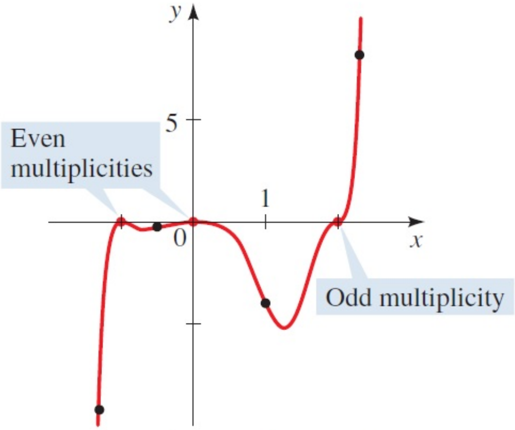
$$P(x) = x^4(x-2)^3(x+1)^2 = (x-0)^4(x-2)^3(x-(-1))^2.$$

We see that $P(x)$ has three real roots $0, 2$, and -1 with multiplicities $4, 3, 2$ respectively.

Also, the leading term of $P(x)$ is x^9, which has a positive coefficient and an odd power, we deduce that $P(x) \to -\infty$ as $x \to -\infty$ and $P(x) \to \infty$ as $x \to \infty$.

With the above information on the x-intercepts and end behavior, we can graph $P(x)$.

Using zeros to graph polynomials - example



$$P(x) = x^4(x-2)^3(x+1)^2$$

Local maxima and minima of polynomials

In the previous example, you might ask how did we know the shape of the graph between two roots? More precisely, how did we know there was only one local minimum between -1 and 0 and similarly in between 0 and 2? You will be able to answer this once you have learned differential calculus.

In general, the derivative of a function tells us the slope of its graph at every point. In particular, the slope is zero at local maxima and local minima of the function. This is how derivatives can help us to sketch graphs of functions with accurate details. At this juncture, we are content with the following theorem (which is proven using differentiation).

Theorem

A degree n polynomial $P(x)$ has at most $n-1$ local extrema.

Activity 2

Graph the following polynomials from the previous activity.

(a) $P(x) = x^4 + x^3 - 16x^2 - 4x + 48$

(b) $Q(x) = -x(x-3)(x+2)$

(c) $R(x) = -(x+1)^2(x-1)^3(x-2)$
Hint: $R(x)$ has only three local extrema.

Activity 2: solution

(a) We factorize $P(x) = x^4 + x^3 - 16x^2 - 4x + 48$ first using the rational zeros theorem. The candidates for the rational roots are $\pm 48, \pm 24, \pm 16, \pm 12, \pm 8, \pm 6, \pm 4, \pm 3, \pm 2, \pm 1$. After trying these values, we find that $-4, -2, 2, 3$ are the roots of $P(x)$. Thus, we have

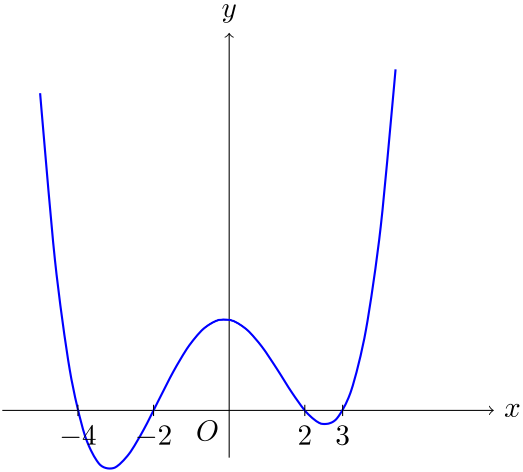
$$P(x) = x^4 + x^3 - 16x^2 - 4x + 48 = (x+4)(x+2)(x-2)(x-3).$$

Note that we get the factorization without long division by comparing the leading coefficients and the degrees of the polynomials.

With these four real roots of multiplicity 1 and the end behavior obtained from the previous activity, there's only one way to sketch the graph with at most $4 - 1 = 3$ local extrema.

Activity 2: solution

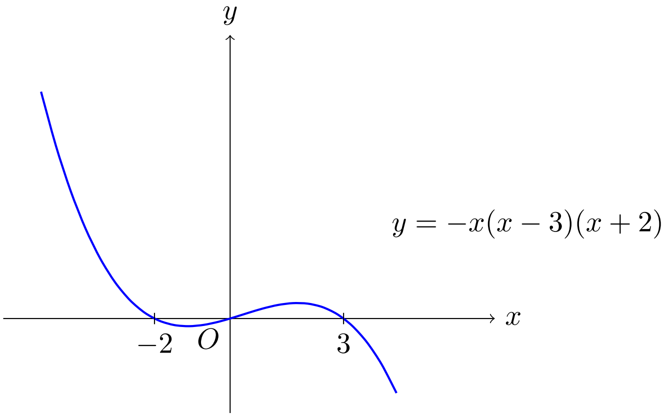
(a)



$$y = (x+4)(x+2)(x-2)(x-3)$$

Activity 2: solution

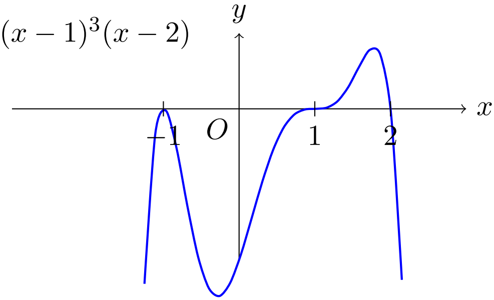
(b) Since $Q(x) = -x(x-3)(x+2)$, it has three real roots of multiplicity 1. Together with the end behavior obtained from the previous activity, there's only one way to sketch the graph with at most $3-1=2$ local extrema.



Activity 2: solution

(c) Since $R(x) = -(x+1)^2(x-1)^3(x-2)$, it has three real roots $-1, 1,$ and 2 of multiplicities $2, 3$ and 1 respectively. With the end behavior obtained from the previous activity, there are a few possibility to sketch the graph with at most $6-1=5$ local extrema. By differential calculus, we can show that $R(x)$ has only three local extrema and hence its graph must look like he following:

$y = -(x+1)^2(x-1)^3(x-2)$



Outline

Rational functions

Definition:

A ***rational function*** is a function of the form

$$f(x) = \frac{P(x)}{Q(x)},$$

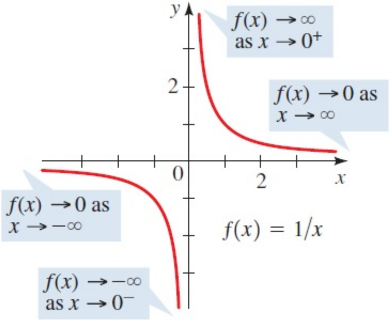
where $P(x), Q(x)$ are polynomials with $Q(x)$ not being the zero constant polynomial.

Remark: The domain of a rational function $f(x) = \frac{P(x)}{Q(x)}$ is $\{x \in \mathbb{R} \mid Q(x) \neq 0\}$.

Even though rational functions are constructed from polynomials, their graphs look quite different from the graphs of polynomial functions.

Graphs of rational functions - example

E.g.: The following is the graph of $f(x) = \frac{1}{x}$. We have seen this last week under transformations of functions.



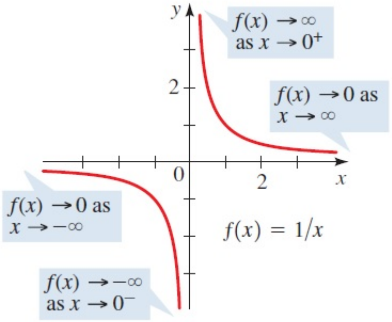
We observe that the value $f(x) = \frac{1}{x}$ increases without bound/gets arbitrarily large when x approaches 0 from the right. We denote this as

$$f(x) \to \infty \quad \text{as} \quad x \to 0^{+}.$$

On the other hand, the value $f(x) = \frac{1}{x}$ decreases without bound/gets arbitrarily negatively large when x approaches 0 from the left. We denote this as

$$f(x) \to -\infty \quad \text{as} \quad x \to 0^{-}.$$

Graphs of rational functions - example



Also, the value $f(x) = \frac{1}{x}$ approaches 0 when $x \to -\infty$ or $x \to \infty$. We denote this as

$$f(x) \to 0 \quad \text{as} \quad x \to -\infty \quad \text{or} \quad x \to \infty.$$

These are the asymptotic behaviors of the rational function $f(x) = \frac{1}{x}$, which are similar for a general rational function.

Vertical asymptotes

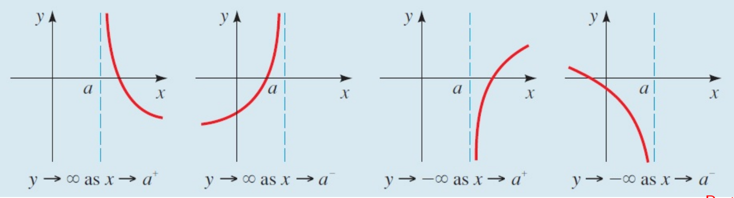
Definition:

Let a be a (finite) real number.

- We say that x ***approaches*** a ***from the right***, denoted as $x \to a^+$, when x gets arbitrarily close to a and $x>a$.
- We say that x ***approaches*** a ***from the left***, denoted as $x \to a^-$, when x gets arbitrarily close to a and $x<a$.

Definition:

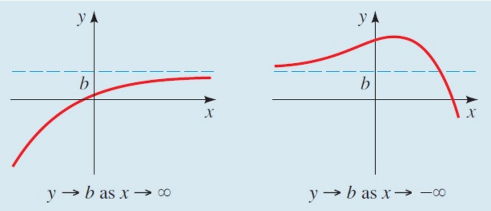
The vertical line $x = a$ is a ***vertical asymptote*** of the function $y = f(x)$ if y approaches ∞ or $-\infty$ as $x \to a^+$ or $x \to a^-$.



Horizontal asymptotes

Definition:

The horizontal line $y = b$ is a ***horizontal asymptote*** of the function $y = f(x)$ if y approaches the finite value b as $x \to -\infty$ or $x \to \infty$.



Finding asymptotes of rational functions

Theorem

Let $f(x)$ be a rational function

$$f(x) = \frac{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0},$$

where the numerator and denominator have no common factors.

1. The vertical asymptotes of f are the lines $x = c$, where c is a zero of the denominator.
2. - If $n < m$, then f has horizontal asymptote $y = 0$.
   - If $n = m$, then f has horizontal asymptote $y = \frac{a_n}{b_m}$.
   - If $n > m$, then f has no horizontal asymptote.

When $n < m$, f is said to be ***proper***, otherwise f is ***improper***.

You don't need to memorize the results of this theorem, they will become intuitive after you have learned limits, which you'll see in Week 5.

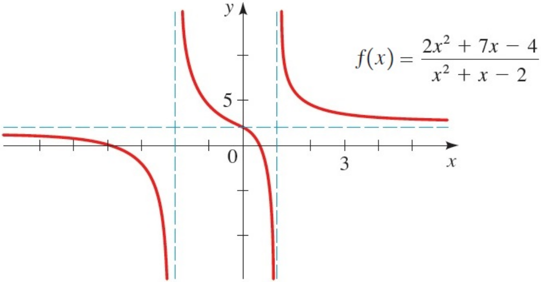
Finding asymptotes of rational functions - example

E.g.: Consider the rational function

$$f(x) = \frac{2x^2 + 7x - 4}{1x^2 + x - 2} = \frac{(2x-1)(x+4)}{(x-1)(x+2)}.$$

By the theorem, we conclude that

1. The vertical asymptotes of f are the lines $x = -2$ and $x = 1$.
2. Since $n = m = 2$, f has horizontal asymptote $y = \frac{2}{1} = 2$.



Activity 3

Find all the x-intercepts and the vertical and horizontal asymptotes of the following rational functions.

(a) $f(x) = \dfrac{2x-3}{x^2-1}$

(b) $g(x) = \dfrac{8x^2+1}{4x^2+2x-6}$

(c) $h(x) = \dfrac{x^3+3x^2}{x^2-4}$

Remark: Similar to polynomials, we need differential calculus to determine the shape of the graph of a rational function for x's that are not large. The bound on the number of local extrema of rational functions is more complicated and it's only enlightening after you have learned differential calculus.

Activity 3: solution

(a) $f(x) = \dfrac{2x-3}{x^2-1} = \dfrac{2x-3}{(x-1)(x+1)}$. The x-intercept is the real root of $2x-3$, which is $\frac{3}{2}$.

1. The vertical asymptotes of f are the lines $x = -1$ and $x = 1$.
2. Since $n = 1 < 2 = m$, f has a horizontal asymptote $y = 0$.

(b) $g(x) = \dfrac{8x^2+1}{4x^2+2x-6} = \dfrac{8x^2+1}{2(2x+3)(x-1)}$. Since $8x^2+1$ has no real roots, g has no x-intercepts.

1. The vertical asymptotes of g are the lines $x = -\frac{3}{2}$ and $x = 1$.
2. Since $n = m = 2$, f has horizontal asymptote $y = \frac{8}{4} = 2$.

Activity 3: solution

(c) $h(x) = \dfrac{x^3 + 3x^2}{x^2 - 4} = \dfrac{x^2(x+3)}{(x+2)(x-2)}$. The x-intercepts are the real roots of $x^2(x+3)$, which are 0 and -3.

1. The vertical asymptotes of h are the lines $x = -2$ and $x = 2$.
2. Since $n = 3 > 2 = m$, f has no horizontal asymptotes.

Outline

Reducing rational inequalities to polynomial inequalities

Last week, we have learned how to use the graphs of functions to solve inequalities. Hence, knowing how to sketch the graphs of polynomial and rational functions enable us to solve rational inequalities.

In fact, we can always reduce a rational inequality to just a polynomial inequality. Thus, knowing how to sketch the graphs of polynomials is enough. This is because

$$\frac{P(x)}{D(x)} \leq 0 \quad \Leftrightarrow \quad \frac{P(x)D(x)}{D(x)^2} \leq 0,$$

and $D(x)^2 > 0$. So the inequality is equivalent to solving

$$P(x)D(x) \leq 0, \quad \text{with} \quad D(x) \neq 0.$$

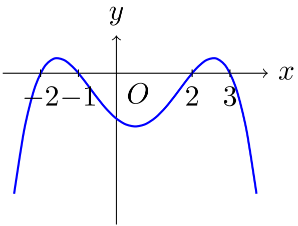
Reducing rational inequalities to polynomial inequalities - example

E.g.: Suppose we want to solve

$$\frac{(2-x)(2+x)}{(x-3)(x+1)} \geq 0 \quad \Leftrightarrow \quad \frac{(2-x)(2+x)(x-3)(x+1)}{(x-3)^2(x+1)^2} \geq 0.$$

Thus, it's equivalent to solve

$$(2-x)(2+x)(x-3)(x+1) \geq 0 \quad \text{and} \quad x \neq 3, -1.$$



$y = (2-x)(2+x)(x-3)(x+1)$

We look for the x's such that the graph of $y = (2-x)(2+x)(x-3)(x+1)$ is above or on the x-axis, excluding the points $x = 3, -1$. Thus, the solution set is $[-2,-1) \cup [2,3)$

Activity 4

Solve the following inequalities by the graphical method.

(a) $\dfrac{1}{x} + \dfrac{1}{1+x} \leq \dfrac{2}{x+2}$

(b) $\dfrac{x^2 + 2x - 3}{3x^2 - 7x - 6} > 0$

(c) $\dfrac{x^3 + 3x^2 - 9x - 27}{x^2 + x + 1} \geq 0$

Activity 4: solution

(a)

$$\frac{1}{x}+\frac{1}{1+x}\leq\frac{2}{x+2} \quad\Leftrightarrow\quad \frac{1+x+x}{x(1+x)}-\frac{2}{x+2}\leq 0$$

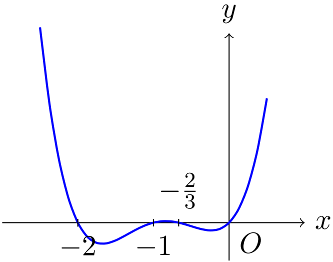
$$\Leftrightarrow\quad \frac{(1+2x)(x+2)-2x(1+x)}{x(1+x)(x+2)}\leq 0 \quad\Leftrightarrow\quad \frac{2+3x}{x(1+x)(x+2)}\leq 0$$

$$\Leftrightarrow\quad \frac{(2+3x)x(1+x)(x+2)}{x^2(1+x)^2(x+2)^2}\leq 0$$

$\Leftrightarrow\ x(2+3x)(1+x)(x+2)\leq 0$ and $x\neq 0,-1,-2.$

Activity 4: solution

(a) $x(2+3x)(1+x)(x+2) \le 0$ and $x \ne 0, -1, -2.$



$y = x(2+3x)(1+x)(x+2)$

We look for the x's such that the graph of $y = (2-x)(2+x)(x-3)(x+1)$ is below or on the x-axis, excluding the points $x = 0, -1, -2$. Thus, the solution set is $(-2,-1) \cup \left[-\frac{2}{3}, 0\right)$.

Activity 4: solution

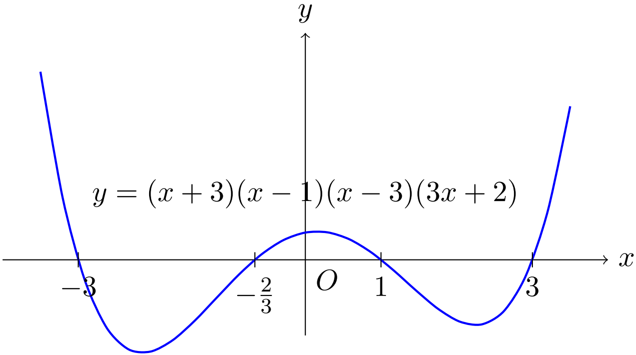
(b)

$$\begin{aligned} & \frac{x^2+2x-3}{3x^2-7x-6} > 0 \\ \Leftrightarrow \quad & \frac{(x+3)(x-1)}{(x-3)(3x+2)} > 0 \\ \Leftrightarrow \quad & \frac{(x+3)(x-1)(x-3)(3x+2)}{(x-3)^2(3x+2)^2} > 0 \end{aligned}$$

$$\Leftrightarrow (x+3)(x-1)(x-3)(3x+2) > 0 \quad \text{and} \quad x \neq -\frac{2}{3}, 3.$$

Activity 4: solution

(b) $(x+3)(x-1)(x-3)(3x+2) > 0$ and $x \neq -\frac{2}{3}, 3$.



We look for the x's such that the graph of $y = (2-x)(2+x)(x-3)(x+1)$ is above the x-axis, excluding the points $x = -\frac{2}{3}, 3$. Thus, the solution set is $(-\infty, -3) \cup \left(-\frac{2}{3}, 1\right) \cup (3, \infty)$.

Activity 4: solution

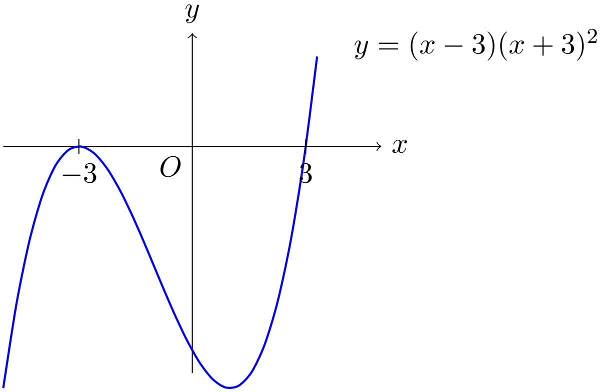
(c) We apply the rational zeros theorem to factorize the numerator. Also, we complete the square for the denominator and see that it's always positive.

$$\begin{aligned} & \frac{x^3+3x^2-9x-27}{x^2+x+1} \geq 0 \\ \Leftrightarrow \quad & \frac{(x-3)(x+3)^2}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}} \geq 0 \end{aligned}$$

$$\Leftrightarrow (x-3)(x+3)^2 \geq 0.$$

Activity 4: solution

(c) $(x-3)(x+3)^2 \geq 0$



We look for the x's such that the graph of $y = (2-x)(2+x)(x-3)(x+1)$ is above or on the x-axis. Thus, the solutions are $x = -3$ or $x \geq 3$.

Outline

Summary

- Graphs of Polynomial Functions
  - graphs of x^n and their end behavior.
  - using zeros to graph polynomials, shape of the graph near a zero.
  - local maxima and minima of polynomials.
- Rational Functions
  - rational functions and their domains.
  - graphs of rational functions.
  - vertical and horizontal asymptotes.
  - finding asymptotes of rational functions.
- Polynomial and Rational Inequalities
  - reducing rational inequalities to polynomial inequalities.

Resources

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 3.2, 3.6, 3.7.